

Xianghui Yang, Ph.D.

✉ Email

in LinkedIn

🔍 Google Scholar

🐙 Github

🌐 Homepage

Employment History

2024.05 – Present 📌 **Senior Applied Scientist (Full-time), Tencent Hunyuan3D, China**

Native 3D generative foundation models: shape generation, mesh generation and 3D VAE representation learning.

- First author of **Hunyuan3D 1.0**, Tencent's first open-source 3D generative system; core author across the **2.0 / 2.1 / 2.5 / Omni** releases and **HunyuanWorld 1.0**.
- Series adopted by the open-source community at scale: **34k+** GitHub stars and **~4m** cumulative downloads on Hugging Face.
- Scaled shape-generation training to **billion-parameter** flow-based diffusion transformers on multi-node GPU clusters, and designed sparse/vertex-latent 3D VAEs that cut latent length while raising reconstruction fidelity.
- Published **17** papers with the team since joining, at CVPR, ICCV, ICML, NeurIPS and AAAI; mentored junior researchers and interns on several of them.

2023.05 – 2023.11 📌 **Applied Scientist Intern (Full-time), Amazon, Australia**

Single-view 3D reconstruction and generation via diffusion models.

- Proposed **ViewFusion**, a *training-free* interpolated-denoising scheme that extends single-view-conditioned diffusion models to multi-view conditioning with no fine-tuning.
- Improved multi-view consistency and reconstruction quality over state-of-the-art baselines on PSNR / SSIM / LPIPS across benchmark datasets.
- Work published as first author at **CVPR 2024**; code released publicly.

Education

2019 – 2023 📌 **Ph.D., The University of Sydney, Australia.**

Supervisors: Dr. Luping Zhou, Dr. Guosheng Lin and Dr. Wanli Ouyang

Thesis title: *Enhancing Novel-class Generalization of Deep Learning Models for Vision Tasks.*

2015 – 2019 📌 **B.Sc. in Physics, Nanjing University, China.**

Thesis title: *Application of Deep Learning in Medical Imaging.*

Research Publications

Google Scholar (Aug. 2026): **1166** citations, *h-index* **15**, *i10-index* **19**. Core author of the **Hunyuan3D** series — among the most widely adopted open-source 3D generative models, with **34k+** cumulative GitHub stars.

First-author Publications

- [1] **Neural Vector Fields: Generalizing Distance Vector Fields by Codebooks and Zero-curl Regularization.** X. Yang, G. Lin, Z. Chen, and L. Zhou. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- [2] **Hunyuan3D 1.0: A Unified Framework for Text-to-3D and Image-to-3D Generation.** X. Yang, H. Shi, B. Zhang, F. Yang, J. Wang, H. Zhao, X. Liu, X. Wang, Q. Lin, J. Yu, et al. *arXiv preprint arXiv:2411.02293*, 2024.
- [3] **ViewFusion: Towards Multi-View Consistency via Interpolated Denoising.** X. Yang, Y. Zuo, S. Ramasinghe, L. Bazzani, G. Avraham, and A. van den Hengel. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [4] **Neural Vector Fields: Implicit Representation by Explicit Learning.** X. Yang, G. Lin, Z. Chen, and L. Zhou. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16 727–16 738, 2023.
- [5] **Single-View 3D Mesh Reconstruction for Seen and Unseen Categories.** X. Yang, G. Lin, and L. Zhou. *IEEE Transactions on Image Processing*, vol. 32, pp. 3746–3758, 2023, DOI: 10.1109/TIP.2023.3279661.
- [6] **BriNet: Towards Bridging the Intra-class and Inter-class Gaps in One-Shot Segmentation.** X. Yang, B. Wang, K. Chen, X. Zhou, S. Yi, W. Ouyang, and L. Zhou. *Proceedings of the 31st British Machine Vision Conference (BMVC)*, 2020, Oral.

Co-authored Publications

- [1] **Mesh-RFT: Enhancing Mesh Generation via Fine-grained Reinforcement Fine-Tuning.** J. Liu, J. Xu, S. Guo, J. Li, J. Yu, H. Weng, B. Lei, X. Yang, Z. Chen, F. Zhu, et al. *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 38, pp. 156 033–156 058, 2026.
- [2] **HY3D-Bench: Generation of 3D Assets.** Team Hunyuan3D, B. Zhang, C. Guo, D. Guo, H. Liu, H. Yan, H. Shi, J. Yu, J. Xu, J. Huang, X. Yang, et al. *arXiv preprint arXiv:2602.03907*, 2026.
- [3] **ArtLLM: Generating Articulated Assets via 3D LLM.** P. Wang, S. Xie, H. Yan, X. Yang, J. Huang, C. Guo, and J. Gu. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [4] **PBR3DGen: A VLM-guided Mesh Generation with High-quality PBR Texture.** X. Wei, B. Zhang, X. Yang, Y. Wang, C. Guo, X. Zhao, and Y. Luximon. *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 13, pp. 10 575–10 583, 2026.
- [5] **Beyond Global Latents: Chunk-Based Sparse Grid VAE for Scalable 3D Modeling.** K. Zhang, Z. Liang, H. Liu, Q. Lin, Z. Lai, Y. Zhao, B. Zhang, X. Yang, Z. Zhao, C. Guo, and L. Quan. *arXiv preprint arXiv:2608.02016*, 2026.
- [6] **HunyuanWorld 1.0: Generating Immersive, Explorable, and Interactive 3D Worlds from Words or Pixels.** HunyuanWorld Team, Z. Wang, Y. Liu, J. Wu, Z. Gu, H. Wang, X. Zuo, T. Huang, W. Li, S. Zhang, X. Yang, et al. *arXiv preprint arXiv:2507.21809*, 2025.
- [7] **TimeFormer: Capturing Temporal Relationships of Deformable 3D Gaussians for Robust Reconstruction.** D. Jiang, Z. Hou, Z. Ke, X. Yang, X. Zhou, and T. Qiu. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 8721–8732, 2025.
- [8] **FlexiTex: Enhancing Texture Generation via Visual Guidance.** D. Jiang, X. Yang, Z. Zhao, S. Zhang, J. Yu, Z. Lai, S. Yang, C. Guo, X. Zhou, and Z. Ke. *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, 4, pp. 3967–3975, 2025.

- [9] **Hunyuan3D 2.5: Towards High-Fidelity 3D Assets Generation with Ultimate Details.** Z. Lai, Y. Zhao, H. Liu, Z. Zhao, Q. Lin, H. Shi, X. Yang, M. Yang, S. Yang, Y. Feng, et al. *arXiv preprint arXiv:2506.16504*, 2025.
- [10] **Unleashing Vecset Diffusion Model for Fast Shape Generation.** Z. Lai, Y. Zhao, Z. Zhao, H. Liu, F. Wang, H. Shi, X. Yang, Q. Lin, J. Huang, Y. Liu, et al. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 2523–2533, 2025.
- [11] **Hunyuan3D Studio: End-to-End AI Pipeline for Game-Ready 3D Asset Generation.** B. Lei, Y. Li, X. Liu, S. Yang, L. Xu, J. Huang, R. Tang, H. Weng, J. Liu, J. Xu, X. Yang, et al. *arXiv preprint arXiv:2509.12815*, 2025.
- [12] **FreeMesh: Boosting Mesh Generation with Coordinates Merging.** J. Liu, H. Weng, B. Lei, X. Yang, Z. Zhao, Z. Chen, S. Guo, T. Han, and C. Guo. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.
- [13] **Hunyuan3D 2.1: From Images to High-Fidelity 3D Assets with Production-Ready PBR Material.** Team Hunyuan3D, S. Yang, M. Yang, Y. Feng, X. Huang, S. Zhang, Z. He, D. Luo, H. Liu, Y. Zhao, Q. Lin, Z. Lai, X. Yang, et al. *arXiv preprint arXiv:2506.15442*, 2025.
- [14] **Hunyuan3D-Omni: A Unified Framework for Controllable Generation of 3D Assets.** Team Hunyuan3D, B. Zhang, C. Guo, H. Liu, H. Yan, H. Shi, J. Huang, J. Yu, K. Li, P. Wang, X. Yang, et al. *arXiv preprint arXiv:2509.21245*, 2025.
- [15] **Nautilus: Locality-aware Autoencoder for Scalable Mesh Generation.** Y. Wang, X. Yi, H. Weng, Q. Xu, X. Wei, X. Yang, C. Guo, L. Chen, and H. Zhang. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 1–10, 2025.
- [16] **Scaling Mesh Generation via Compressive Tokenization.** H. Weng, Z. Zhao, B. Lei, X. Yang, J. Liu, Z. Lai, Z. Chen, Y. Liu, J. Jiang, C. Guo, et al. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11 093–11 103, 2025.
- [17] **Hunyuan3D 2.0: Scaling Diffusion Models for High Resolution Textured 3D Assets Generation.** Z. Zhao, Z. Lai, Q. Lin, Y. Zhao, H. Liu, S. Yang, Y. Feng, M. Yang, S. Zhang, X. Yang, et al. *arXiv preprint arXiv:2501.12202*, 2025.

Selected Open-source Projects

- | | |
|----------------|---|
| 2024 – Present | <ul style="list-style-type: none"> Hunyuan3D Series — github.com/Tencent-Hunyuan
 Open-source 3D generative foundation models (1.0 / 2.0 / 2.1 / 2.5 / Omni / World). Contributor as first or core author. Cumulatively 34k+ GitHub stars and ~4m model downloads on Hugging Face. |
| 2024 | <ul style="list-style-type: none"> Hunyuan3D 1.0 — Tencent-Hunyuan/Hunyuan3D-1
 First author. Tencent’s first open-source 3D generative system, unifying text-to-3D and image-to-3D generation. 3.5k stars. ViewFusion — Wi-sc/ViewFusion
 First author, CVPR 2024. Training-free multi-view consistent diffusion via interpolated denoising. |
| 2023 | <ul style="list-style-type: none"> NVF — Wi-sc/NVF
 First author, CVPR 2023 & TPAMI 2025. Official implementation of Neural Vector Fields, an implicit–explicit surface representation. GenMesh — Wi-sc/GenMesh
 First author, IEEE TIP 2023. Single-view 3D mesh reconstruction generalizing to unseen object categories. |

Academic Service

- Conferences  **Reviewer.** CVPR, ICCV, ECCV, SIGGRAPH Asia, AAAI, ACM Multimedia, BMVC
- Journals  **Reviewer.** IEEE Transactions on Image Processing (TIP), Pattern Recognition

Miscellaneous Experience

Awards and Achievements

- 2023  **Paulette Isabel Jones Career Award.** Awarded by The University of Sydney
- 2020  **Faculty of Engineering Research Scholarship.** Awarded by The University of Sydney
- 2019  **Outstanding Graduates Award.** Awarded by Nanjing University
- 2017  **MCM/ACM Honorable Mention.** Awarded by the Consortium for Mathematics and Its Application
- 2016 – 2018  **Outstanding Student Award (×2), People's Scholarship (×4) and Xingquan Responsibility Scholarship.** Awarded by Nanjing University

Certification

- 2022  **Financial Risk Manager (FRM), Part I.** Awarded by the Global Association of Risk Professionals.